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Photoredox copper-catalyzed regioselective syntheses of allenols *via* HAT-mediated propargyl radical generation†

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Selectivity control of the reactions of propargylic and/or allenylic radicals remains particularly elusive. Here, a visible light-mediated Cu-catalyzed protocol for the highly selective syntheses of tri- and tetra-substituted allenols from alkynol derivatives has been developed. The synthetic potential of the products has been demonstrated. The control experiments, Stern–Volmer quenching, and cyclic voltammetry studies support a reaction pathway involving the generation of allenyl radicals *via* photo-mediated N–O bond scission, an intramolecular 1,5-HAT process, the resonance to propargylic radicals, and coupling. A non-common intramolecular 1,6-HAT process has also been observed.

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Introduction

Selectivity control is an everlasting topic in organic synthesis, advanced materials, and medicinal chemistry or life science.[1] For example, it is of particular interest to see the chemistry on the propargylic/allenylic species due to the versatile potential of alkynes and allenes. Recently, much attention has been paid to the generation and reactivity of propargylic and/or allenylic radicals affording alkynes[2] and/or allenes.[3–6] However, the efficient generation of propargylic radicals and the selectivity-controlled trapping still remain particularly elusive (Fig. 1a). 1,5-Hydrogen atom transfer (HAT)[7,8] mediated by alkoxyl radicals[9,10] has emerged as an important strategy for C(sp^3)–H bond functionalization (Fig. 1b). Given the high efficiency of radical formation with this specific method, we wonder whether such an alkoxyl radical-induced HAT of alkynes would efficiently generate propargylic radicals followed by transition metal-catalyzed functionalization, providing an efficient and selective platform for synthesizing multi-substituted alkynes or allenes (Fig. 1c). However, notable challenges for such a concept included (Fig. 1c): firstly, the tautomeric equilibrium of propargylic radicals to allenyl radicals and the resulting control of selectivity for trapping of these two radicals;[2–4] secondly, the introduction of the alkyne functional group and the formation of the allene functional group may lead to direct cyclic radical addition reactions;[11] and thirdly, the facile HAT of allenyl radicals could produce other non-functionalized allene by-products before the transition metal-

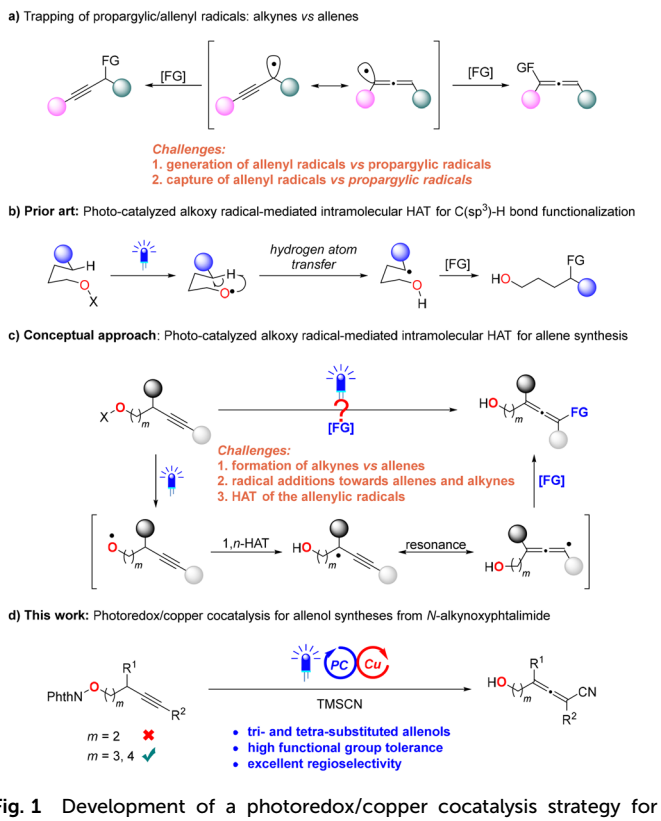


Fig. 1 Development of a photoredox/copper cocatalysis strategy for allenol syntheses based on 1,5-HAT.

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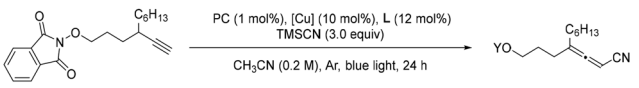
catalyzed functionalization occurs. In addition, it would be a platform reaction for the study on how far a HAT reaction may go. Here, we report our success on the exclusive synthesis of tri- and tetra-substituted allenes *via* visible-light photoredox and copper dual catalysis (Fig. 1d).

Results and discussion

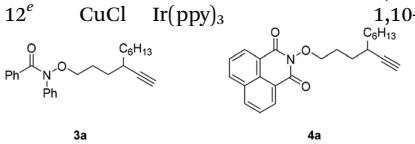
Our initial attempt used *N*-alkynoxyphthalimide **1a**, which is easily obtained by the Mitsunobu reaction from the corresponding alkynol. We began investigating the coupling reaction of **1a** with trimethylsilyl cyanide by applying a catalytic system consisting of 4CzIPN and CuCl/dmbpy in CH_3CN under blue LED irradiation, and the desired allene product **2a′** was afforded in 38% yield as determined by ^{1}H NMR spectroscopy (Table 1, entry 1), without the formation of an alkyne product. The photocatalyst MesAcr$^+$ClO$_4^-$ resulted in a largely decreased yield of the allene product **2a′** (Table 1, entry 2). Further exploration of photocatalysts revealed that iridium complexes, such as Ir[(dtbbpy)(ppy)$_2$]PF$_6$ or Ir(ppy)$_3$, were more efficient (Table 1, entries 3 and 4). Other copper catalysts such as CuBr and CuI (Table 1, entries 5 and 6) were also screened and CuCl was still the best. Subsequently, we tested a series of other bipyridine ligands, and identified 1,10-phenanthroline as equally excellent as dmbpy (Table 1, entry 7). We chose 1,10-phen as the ligand of preference finally in consideration of the price. Other alkynoxylamide derivatives, such as **3a** and **4a**, exhibited a much lower reactivity (Table 1, entries 8 and 9). Control experiments demonstrated that the CuCl/1,10-phen system, photocatalyst and light are indispensable for enabling this transformation to proceed smoothly (entries 10–12).

With the optimized conditions in hand, we conducted further investigations on the substrate scope of this reaction (Fig. 2). We found that a wide range of terminal alkynols are suitable for this transformation, leading to the successful preparation of the corresponding tri-substituted allenols **2a–2t** smoothly. In terms of the alkyl groups R^1 at the propargyl position, non-functionalized linear primary alkyl chains of various lengths (**2a–2d**) as well as alkyl substituents bearing ethers (**2e** and **2f**), chloro (**2g**), acetal (**2h**), terminal olefin (**2i**), and aryls

Table 1 Optimization of the reaction conditions[a]



Entry	[Cu]	PC	Ligand	Yields of 2a′/2a[b] (%)	RSM[b] (%)
1	CuCl	4CzIPN	dmbpy	38/0	50
2	CuCl	MesAcr$^+$ClO$_4^-$	dmbpy	7/0	78
3	CuCl	Ir[(dtbbpy)(ppy)$_2$]PF$_6$	dmbpy	80/0	0
4	CuCl	Ir(ppy)$_3$	dmbpy	81/0	0
5	CuBr	Ir(ppy)$_3$	dmbpy	79/0	0
6	CuI	Ir(ppy)$_3$	dmbpy	60/0	0
7	CuCl	Ir(ppy)$_3$	1,10-phen	81/0	0
8[c]	CuCl	Ir(ppy)$_3$	1,10-phen	0/0	79
9[d]	CuCl	Ir(ppy)$_3$	1,10-phen	24/0	75
10	—	Ir(ppy)$_3$	—	12/0	2
11	CuCl	—	1,10-phen	0/0	96
12[e]	CuCl	Ir(ppy)$_3$	1,10-phen	0/0	92



[a] The reaction was conducted with **1a** (0.2 mmol), PC (1 mol%), [Cu] (10 mol%), **L** (12 mol%), and TMSCN (0.6 mmol) in CH_3CN (1 mL) under an Ar atmosphere, with blue LED (50 W, 460 nm) irradiation for 24 h. [b] The yield was determined by ^{1}H NMR spectroscopy with CH_2Br_2 as the internal standard. [c] **3a** instead of **1a**. [d] **4a** instead of **1a**. [e] The reaction was conducted without light. dmbpy = 4,4′-dimethyl-2,2′-bipyridine. 1,10-phen = 1,10-phenanthroline.

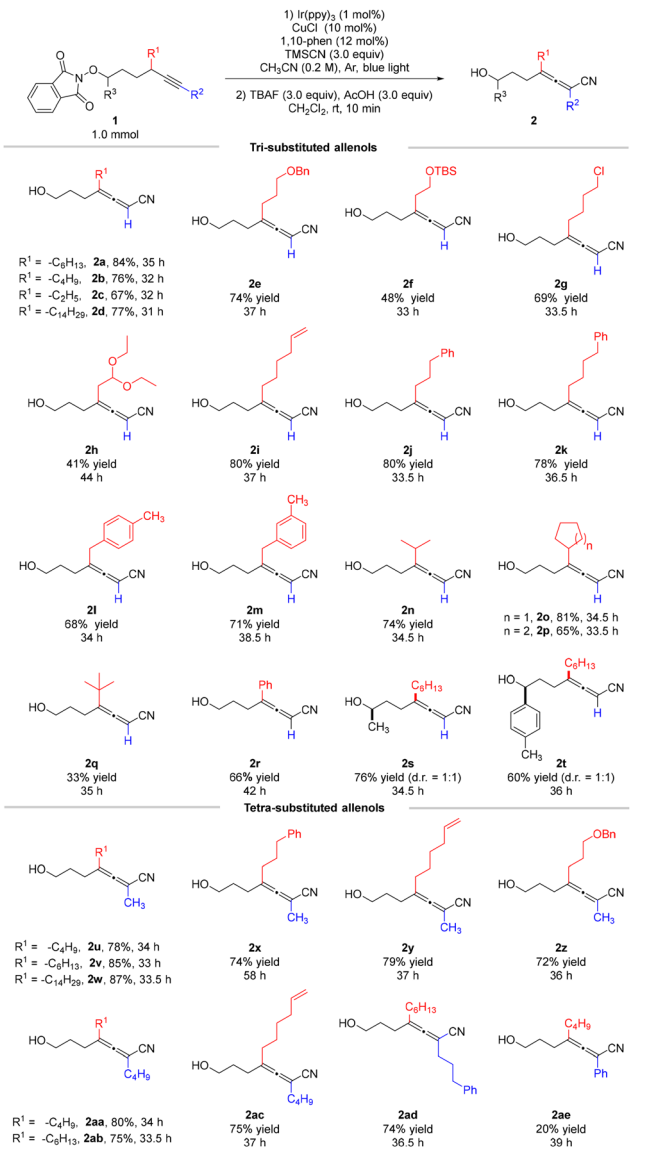


Fig. 2 Substrate scope for the syntheses of tri- and tetra-substituted allenols.

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(**2j**–**2m**) were all found to be compatible under the standard conditions. R^1 may also be secondary alkyl groups (**2n**–**2p**) or even a tertiary alkyl group (**2q**), resulting in the corresponding allenols in modest to good yields. Furthermore, beyond the alkyl-substituted substrates, the reaction of the substrate with R^1 as phenyl yielded **2r** smoothly, achieving a yield of 66%. Furthermore, the synthesis of secondary allenols **2s** and **2t** was realized with yields of 76% and 60%, respectively. Finally, this protocol also worked for internal alkynols with R^2 being alkyl, phenylalkyl or phenyl groups, affording tetra-substituted allenols (**2u**–**2ae**).

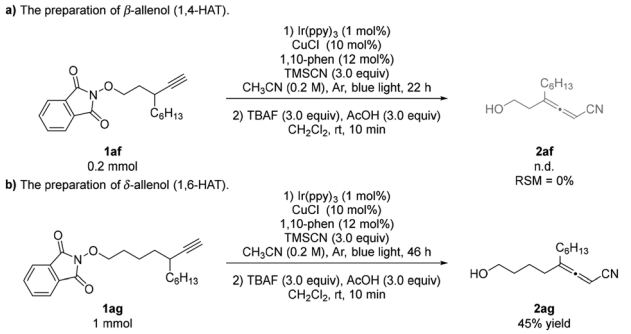


Fig. 3 Investigation of the 1,4- and 1,6-HAT pathways.

Furthermore, various *N*-alkynoxyphthalimide precursors with diverse tether sizes were employed to investigate the potential of our protocol for 1,4- or 1,6-HAT (Fig. 3). When subjected to the standard conditions, compound **1af** failed to yield the corresponding beta-allenol **2af**, with the complete consumption of the starting material observed (Fig. 3a). Conversely, the reaction of substrate **1ag** with TMSCN resulted in the formation of delta-allenol **2ag** in 45% yield, involving 1,6-HAT. Furthermore, endeavors to synthesize the initial *N*-alkynoxyphthalimide precursor for alpha-allenols proved unsuccessful (see Fig. S4 in the ESI†).

To demonstrate the synthetic potential of this reaction, a gram-scale reaction was conducted using 7 mmol of **1a**, resulting in the formation of allenol **2a** with an isolated yield of 73% (1.1 g) (Fig. 4a). These allenols are versatile synthetic building blocks as shown in Fig. 4b. In the presence of Cs_2CO_3, allenol **2a** underwent oxycyclization to give the dihydropyran product **5**, which may further react with 4-chlorobenzyl bromide[12] to produce the dihydropyran product **6** in a one-pot reaction. The

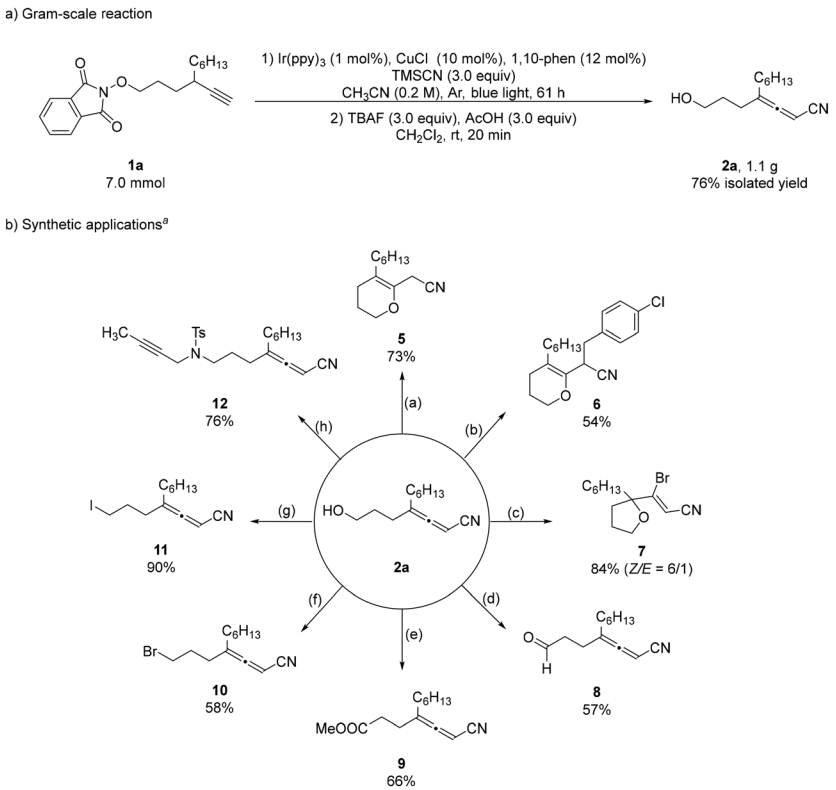
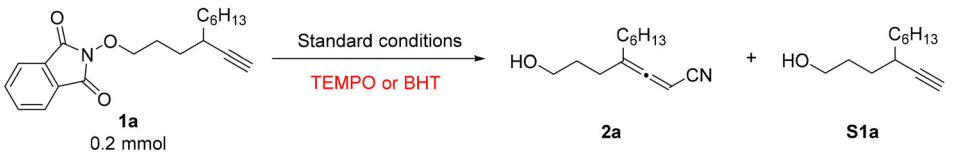


Fig. 4 Gram-scale reaction and synthetic applications. [a] Reaction conditions: (a) **2a** (0.2 mmol), Cs_2CO_3 (1.0 equiv.), CH_3CN, rt, 12 h; (b) **2a** (0.2 mmol), 4-chlorobenzyl bromide (1.1 equiv.), TBAI (5 mol%), NaH (2.0 equiv.), THF, 0 °C to rt, Ar, 22 h; (c) **2a** (0.2 mmol), NBS (1.2 equiv.), DCM/THF = 4/1, rt, 12 h; (d) **2a** (0.2 mmol), $Fe(NO_3)_3{\cdot}9H_2O$ (20 mol%), TEMPO (20 mol%), NaCl (20 mol%), DCE (0.05 M), O_2, rt, 11 h; (e) (i) **2a** (0.2 mmol), $Fe(NO_3)_3{\cdot}9H_2O$ (20 mol%), TEMPO (20 mol%), KCl (20 mol%), DCE (0.2 M), O_2, rt, 30 h; (ii) $TMSCHN_2$ (1.5 equiv.), Et_2O/MeOH = 4/1, 0 °C to rt, 15 h; (f) **2a** (0.2 mmol), PPh_3 (1.5 equiv.), CBr_4 (1.5 equiv.), DCM, 0 °C to rt, 24 h; (g) **2a** (0.2 mmol), I_2 (1.2 equiv.), PPh_3 (1.2 equiv.), imidazole (1.2 equiv.), DCM, rt, 0.5 h; (h) **2a** (1.15 equiv.), *N*-(but-2-yn-1-yl)-4-methylbenzenesulfonamide (0.2 mmol), PPh_3 (1.35 equiv.), DIAD (1.35 equiv.), THF, 0 °C to rt, Ar, 23 h.

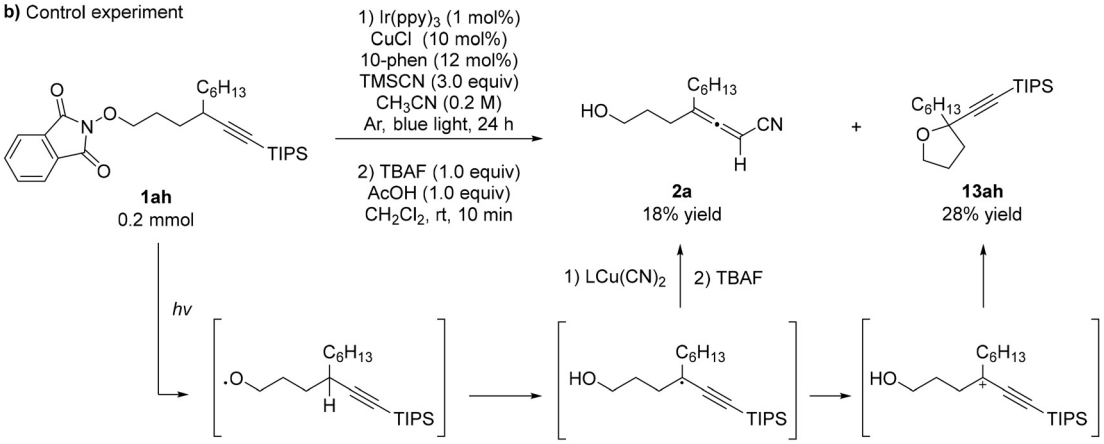
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a) Radical inhibiting experiments

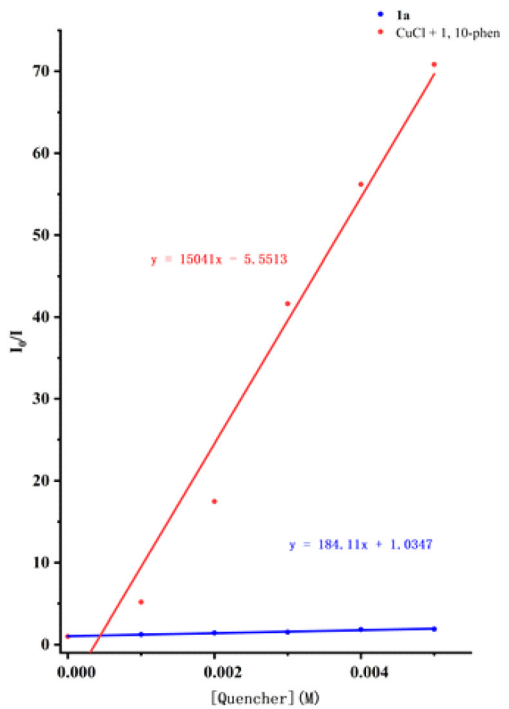


Radical inhibitor	NMR yield of **2a**	Recovery of **1a**	NMR yield of **S1a**
/	81%	0	/
TEMPO (2.0 equiv)	30%	24%	/
TEMPO (5.0 equiv)	2%	75%	/
BHT (2.0 equiv)	38%	13%	18% (18%)[a]
BHT (5.0 equiv)	18%	44%	29%

b) Control experiment



c) Stern-Volmer quenching experiments of Ir(ppy)$_3$



d) Cyclic voltammograms

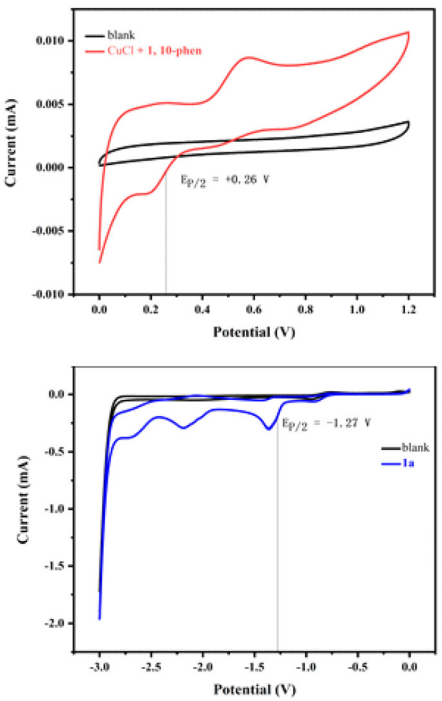


Fig. 5 Mechanistic studies. [a] Isolated yield.

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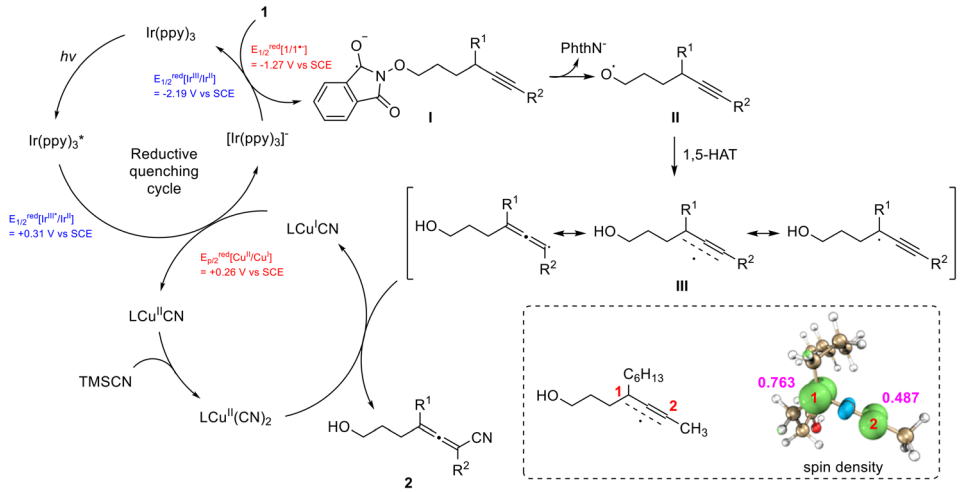


Fig. 6 Proposed mechanism.

tetrahydrofuran product **7** was obtained through bromocyclization[13] with NBS in 84% yield. In addition, aerobic oxidation of **2a** afforded allenal **8** under the catalysis of 20 mol% each of Fe$(NO_3)_3{\cdot}9H_2O$, TEMPO, and NaCl in DCE.[14] Meanwhile, under the catalytic system of Fe$(NO_3)_3{\cdot}9H_2O$, TEMPO, and KCl, **2a** was converted into carboxylic acid, which was further esterified with $TMSCHN_2$ in one pot to provide allenoate **9** in 66% yield over two steps.[15] Upon treatment with CBr_4 and I_2, the allenol product **2a** could be easily transformed into the synthetically versatile allenyl bromide **10** and allenyl iodide **11**, respectively.[16,17] Finally, **2a** may readily undergo a Mitsunobu reaction[18] to afford allenyne **12** in a decent yield.

To gain mechanistic insights into the reaction, firstly, we conducted the reaction of **1a** in the presence of radical trapping agents such as TEMPO and BHT. As shown in Fig. 5a, the yield of the desired allenols dropped dramatically with the increase in the amount of the radical inhibitor. In addition, when BHT was used, the alkynol product **S1a**, resulting from intermolecular HAT of the initially formed alkoxyl radical, was detected. When the internal alkyne **1ah** substituted with the triisopropylsilyl group was subjected to the standard conditions, in addition to the normal allene product **2a**, the isolation of the tetrahydrofuranyl triisopropylsilyl acetylene product **13ah** indicated the involvement of a propargylic cation, which may be formed *via* the oxidation of a propargylic radical intermediate (Fig. 5b). Moreover, Stern–Volmer quenching experiments (Fig. 5c) demonstrated that the CuCl/1,10-phen catalyst, rather than the substrate **1a**, quenched the excited-state photocatalyst Ir$(ppy)_3$, suggesting a reductive quenching pathway. The half peak potential of the catalyst CuCl/1,10-phen $E^{red}_{p/2}$ [Cu^{II}/Cu^{I}] was determined by Cyclic Voltammetry (CV) experiments to be +0.26 V *vs.* SCE in CH_3CN and the half peak potential of substrate **1a** $E^{red}_{p/2}$ [**1a**/**1a**$^{\bullet-}$] was determined to be −1.27 V *vs.* SCE in CH_3CN (Fig. 5d).

On the basis of these experimental results, a plausible reaction mechanism was proposed (Fig. 6). Initially, the excitation of the photocatalyst Ir$(ppy)_3$ leads to its excited state Ir*$(ppy)_3$ ($E^{red}_{1/2}$ [Ir^{III*}/Ir^{II}] = +0.31 V *vs.* SCE in CH_3CN),[19] which can be reductively quenched with $LCu^{I}CN$ ($E^{red}_{p/2}$ [Cu^{II}/Cu^{I}] = +0.26 V *vs.* SCE in CH_3CN) to generate $LCu^{II}(CN)$ and [Ir$(ppy)_3]^-$ species. [Ir$(ppy)_3]^-$ ($E^{red}_{1/2}$ [Ir^{III}/Ir^{II}] = −2.19 V *vs.* SCE in CH_3CN)[19] would reduce substrate **1** ($E^{red}_{p/2}$ [**1a**/**1a**$^{\bullet-}$] = −1.27 V *vs.* SCE), producing radical anion intermediate **I** and regenerating the photocatalyst Ir$(ppy)_3$. Subsequently, alkoxy radical intermediate **II** would be generated *via* the cleavage of the N–O bond of intermediate **I**. The subsequent intramolecular 1,5-HAT would generate radical intermediate **III**. Mulliken atomic spin density analysis suggests that intermediate **III** is a delocalized radical, with the spin density distributed across both ends of the system (0.763 at C1 and 0.487 at C2, with R^1 being *n*-hexyl and R^2 being methyl). Finally, intermediate **III** can be trapped by the $LCu^{II}(CN)_2$ species, leading to the formation of the allene product **2**.

Conclusions

In summary, we have developed a unified platform for the highly selective preparation of tri- and tetra-substituted allenols. This approach leverages dual copper and photoredox catalysis to enable the generation and functionalization of allenyl radicals. This mild reaction demonstrates a broad substrate scope, and the post-reaction transformations of the allenol products highlight the significant synthetic potential of this methodology.

Author contributions

S. M., H. Y. and W. L. designed the experiments. W. L. performed the experiments. W. L., H. Y. and S. M. wrote the manuscript. W. L., H. Y. and S. M. checked the experimental data.

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Data availability

All detailed procedures, characterization data, computational details, and spectra are available in the ESI.†

Conflicts of interest

The authors declare no competing interests.

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